don’t, and some physicists have even suggested experiments to detect the possible influence that such a future constraint might have on things that we can observe today. For an interesting article discussing the possibility of future and past constraints on entropy, see Murray Gell-Mann and James Hartle, “Time Symmetry and Asymmetry in Quantum Mechanics and Quantum Cosmology,” in Physical Origins of Time Asymmetry, J. J. Halliwell, J. Pérez-Mercader, W. H. Zurek, eds. (Cambridge, Eng.: Cambridge University Press, 1996), as well as other papers in Parts 4 and 5 of that collection.

18. Throughout this chapter, we’ve spoken of *the* arrow of time, referring to the apparent fact that there is an asymmetry along the time axis (any observer’s time axis) of spacetime: a huge variety of sequences of events is arrayed in one order along the time axis, but the reverse ordering of such events seldom, if ever, occurs. Over the years, physicists and philosophers have divided these sequences of events into subcategories whose temporal asymmetries might, in principle, be subject to logically independent explanations. For example, heat flows from hot objects to cooler ones, but not from cool objects to hot ones; electromagnetic waves emanate outward from sources like stars and lightbulbs, but seem never to converge inward on such sources; the universe appears to be uniformly expanding, and not contracting; and we remember the past and not the future (these are called the thermodynamic, electromagnetic, cosmological, and psychological arrows of time, respectively). All of these are time-asymmetric phenomena, but they might, in principle, acquire their time asymmetry from completely different physical principles. My view, one that many share (but others don’t), is that except possibly for the cosmological arrow, these temporally asymmetric phenomena are not fundamentally different, and ultimately are subject to the same explanation—the one we’ve described in this chapter. For example, why does electromagnetic radiation travel in expanding outward waves but not contracting inward waves, even though both are perfectly good solutions to Maxwell’s equations of electromagnetism? Well, because our universe has low-entropy, coherent, ordered sources for such outward waves—stars and lightbulbs, to name two—and the existence of these ordered sources derives from the even more ordered environment at the universe’s inception, as discussed in the main text. The psychological arrow of time